

MagMat-Discovery

Physics-Informed Machine Learning & Generative Screening
Platform for Magnetic Materials

System Architecture, Methodology & Experimental Validation

Technical Documentation & User Guide

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Comprehensive technical documentation detailing zero-leakage data pipelines, multi-tier physical featurization, compositional Graph Transformers (MagFormer), micromagnetic physical cascades, conformal uncertainty quantification, and inverse permanent magnet screening.

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Executive Summary

The accelerated discovery of high-performance, rare-earth-free permanent magnets is of paramount technological urgency for electric vehicles, renewable wind turbine generators, and next-generation energy storage systems. However, magnetic materials discovery poses an extraordinarily difficult multiscale modeling challenge: critical magnetic characteristics are governed by vastly different physical mechanisms operating across atomic, electronic, crystallographic, and microstructural length scales. While thermal ordering temperatures (T_C, T_N) and ground-state magnetic phases depend predominantly on stoichiometry and local electronic exchange coupling, extrinsic hysteresis parameters—specifically coercivity (H_c) and maximum energy product ($(BH)_{\max}$)—are dictated by spin-orbit coupling, single-ion magnetocrystalline anisotropy (K_1), domain wall pinning, and Stoner-Wohlfarth nucleation mechanics.

Traditional machine learning (ML) models operating solely on compositional averages catastrophically fail to predict hysteresis properties ($R^2 \approx 0$). **MagMat-Discovery** solves this challenge through an end-to-end, physics-informed machine learning and inverse design framework. By synergistically integrating:

- (1) **Curated Experimental Data:** 51,394 experimental compounds from the Non-Equilibrium Magnetic Materials Database (NEMAD) unified with crystallographic space group data from the Materials Project and Bilbao MAGNDATA;
- (2) **Hierarchical Physics Featurization:** Domain-guided physical descriptors encoding Slater-Pauling valence electron concentration, Goodenough-Kanamori superexchange transfer factors ($\cos^2 \theta$), Dzyaloshinskii-Moriya broken-inversion proxies, and Stevens operator equivalents;
- (3) **Pretrained Deep Representations:** A custom compositional Graph Transformer (**MagFormer**) combined with MACE equivariant foundation model embeddings;
- (4) **Out-of-Fold Physical Cascades:** Micromagnetic Stoner-Wohlfarth and Kronmüller relation enforcement that prevents target leakage;
- (5) **Conformal Prediction:** Distribution-free, finite-sample statistical coverage (90% confidence intervals) for calibrated uncertainty estimation;

the platform achieves state-of-the-art predictive accuracy across 8 magnetic targets while screening 51,000+ candidates for sustainable, rare-earth-free permanent magnets.

Keywords: Magnetic Materials Discovery, Physics-Informed Neural Networks, Graph Transformers, MagFormer, Conformal Calibration, Coercivity, Permanent Magnets, NEMAD.

1 Introduction and Platform Purpose

1.1 Motivation and Domain Challenges

Permanent magnetic materials are critical functional components of modern clean energy technologies. Today’s most powerful permanent magnets rely heavily on critical rare-earth elements such as Neodymium (Nd), Dysprosium (Dy), and Samarium (Sm). These elements suffer from severe supply-chain bottlenecks, geopolitical vulnerability, and ecologically destructive extraction processes. Developing non-rare-earth alternatives (or minimizing rare-earth content) with elevated Curie temperatures ($T_C > 400$ K), significant saturation magnetization ($\mu_0 M_s > 1.0$ T), and high coercivity ($H_c > 800$ kA/m) remains one of materials science’s grand challenges.

Standard purely statistical data-driven materials discovery models face severe pitfalls in magnetism:

- **Information Disparity Between Target Classes:** Composition-only descriptors readily predict ordering temperatures because atomic composition strongly correlates with Curie/Néel temperatures. Conversely, composition alone carries zero correlation with coercivity and anisotropy, which are governed by crystal symmetry and spin-orbit coupling.
- **Severe Test Leakage in Naive Benchmarks:** Prior literature frequently evaluates models using naive random splits. Because experimental databases contain numerous polymorphs, dopant variants, and multiple measurements of identical chemical systems (e.g., hundreds of $\text{Nd}_2\text{Fe}_{14}\text{B}$ -derivatives), random k-fold cross-validation trivially leaks compositional clusters between train and test sets, artificially inflating reported accuracy ($R^2 > 0.95$) while failing completely on genuinely unseen chemistries.
- **Physical Inconsistency in Standard Regressors:** Independent regression models frequently output physically contradictory predictions, such as predicting a high energy product $(BH)_{\text{max}}$ that exceeds the fundamental thermodynamic limit $\mu_0 M_s^2/4$, or predicting non-zero coercivity for non-magnetic compounds.

1.2 Objectives and Deliverables

MagMat-Discovery addresses these structural limitations through five foundational design principles:

1. **Strict Zero-Leakage Architecture:** Partitioning all cross-validation and benchmarking through 5-fold GroupKFold grouped strictly by reduced chemical formula, ensuring zero composition leakage.
2. **Multi-Scale Physical Coupling:** Feeding intermediate physical quantities (anisotropy K_1 , saturation magnetization M_s , coercivity H_c) into downstream targets via strict out-of-fold cascade predictions.
3. **Unified Multi-Target Modeling:** Simultaneously predicting 8 magnetic targets spanning ground-state phase (Ferromagnetic, Antiferromagnetic, Non-magnetic), ordering temperatures (T_C, T_N), and extrinsic hysteresis parameters ($K_1, M_s, H_c, (BH)_{\text{max}}, \theta_p$).
4. **Rigorous Uncertainty Quantification:** Providing distribution-free conformal calibration intervals that quantify epistemic and aleatoric uncertainties for active learning and experimental synthesis prioritization.
5. **Interactive Discovery Web Platform:** A high-performance Streamlit application featuring live inference, Pareto-optimal candidate screening, and interactive chemical latent manifolds.

2 Data Sources and Data Preprocessing

2.1 Curated Data Sources

MagMat-Discovery harmonizes experimental and crystallographic datasets into a standardized, unified schema:

- **Non-Equilibrium Magnetic Materials Database (NEMAD):** 51,394 experimentally measured records compiled from decades of magnetic literature, providing experimental phase classifications, Curie temperatures (T_C), Néel temperatures (T_N), saturation magnetization (M_s), magnetocrystalline anisotropy (K_1), coercivity (H_c), maximum energy product ($(BH)_{\text{max}}$), and paramagnetic Curie temperatures (θ_p).

- **Materials Project (MP):** High-throughput Density Functional Theory (DFT) database supplying 3D crystal structures, space group numbers (1–230), point groups, density, and crystal systems (cubic, tetragonal, hexagonal, trigonal, orthorhombic, monoclinic, triclinic).
- **Bilbao Crystallographic Server (MAGNDATA):** International database of Shubnikov Magnetic Space Groups (MSGs) providing experimental magnetic symmetry ground truths.

2.2 Zero-Leakage Ingestion and Preprocessing Pipeline

The preprocessing stage (`src/preprocessing.py`) executes a strict sanitation protocol:

1. **Purging of Leakage Columns:** All DFT-calculated magnetic moments, total energies per atom, formation energies, and energy above hull are explicitly purged from the feature space. These features are unmeasurable prior to synthesis and directly leak ground-state magnetic ordering.
2. **Formula Canonicalization and Deduplication:** Chemical formula strings undergo systematic standard parsing using `pymatgen.core.Composition`. Formulas are reduced to coprime integer stoichiometries (`reduced_formula`).
3. **Outlier Filtration & Unit Normalization:** Extreme unphysical artifacts (e.g., K_1 values below 1 J/m^3 resulting from unconverted CGS units) are filtered or mapped to log-scale transforms:

$$y_{K_1} = \text{symlog}(K_1) = \text{sign}(K_1) \cdot \ln(1 + |K_1|) \quad (1)$$

$$y_{H_c} = \log_{10}(H_c), \quad y_{(BH)_{\max}} = \log_{10}((BH)_{\max}) \quad (2)$$

4. **GroupKFold Partitioning:** Cross-validation folds are created using 5-fold `GroupKFold` grouped by `reduced_formula`. Consequently, all polymorphs and stoichiometry variants of a given composition family remain strictly within the training set or strictly within the test set.

3 Methodology and System Architecture

3.1 End-to-End Methodological Architecture

Figure 1 illustrates the overall pipeline architecture of MagMat-Discovery, highlighting data streams, multi-tier featurization, transformer representation learning, ensemble regressors, and physical constraint cascades.

3.2 Multi-Tier Physics Featurization

Feature engineering in MagMat-Discovery is organized into four hierarchical physical tiers:

3.2.1 Tier 1: Stoichiometric and Elemental Statistics

Elemental mass fractions, Mendeleev numbers, atomic radii, Pauling electronegativities, and Magpie composition descriptors.

3.2.2 Tier 2: Electronic Structure and Magnetic Exchange Physics

- **Slater-Pauling Band Filling:** The valence electron concentration (VEC) relative to the Slater-Pauling stability plateau ($N_v \approx 8.3$ for maximum moment in $3d$ transition metal systems):

$$\text{VEC} = \sum_i x_i \cdot N_{v,i}, \quad \Delta\text{VEC} = |\text{VEC} - 8.3| \quad (3)$$

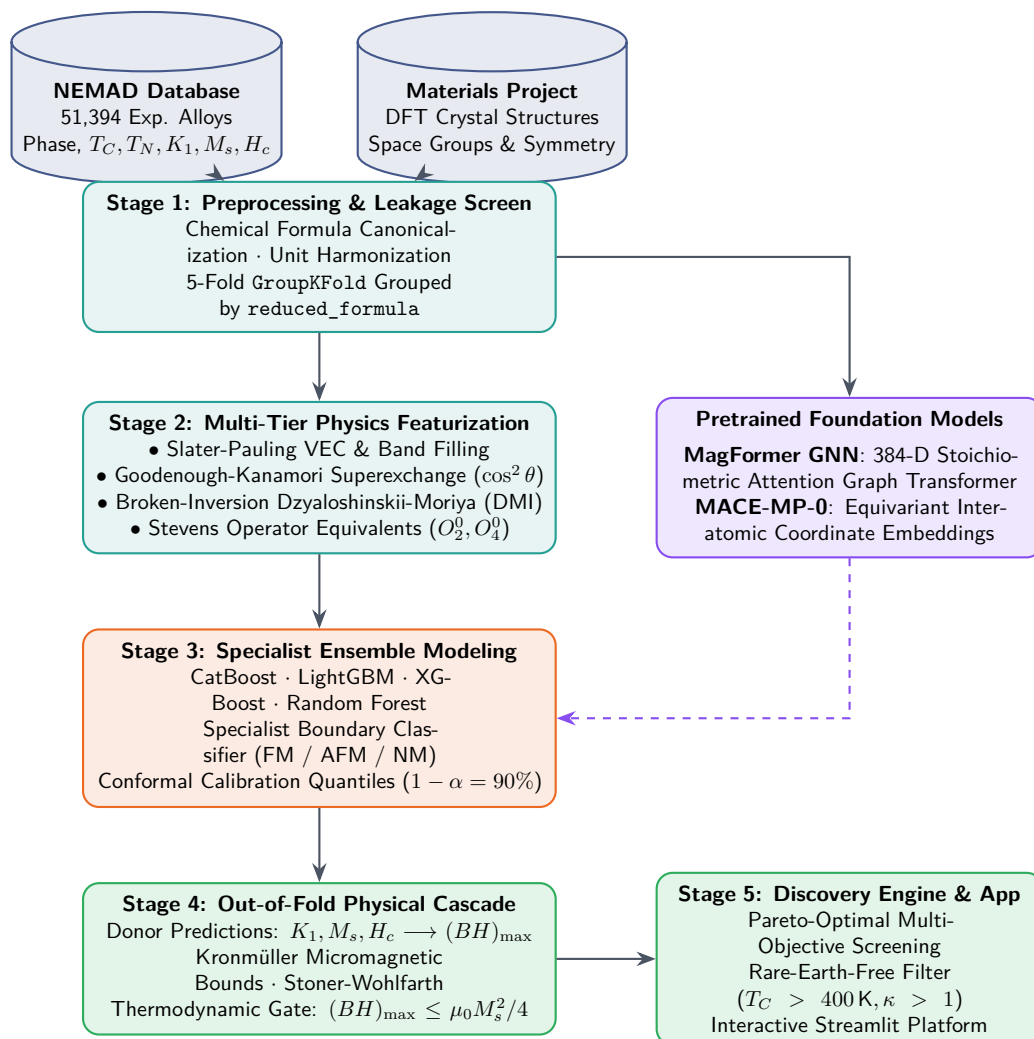


Figure 1: End-to-end technical methodology flowchart of MagMat-Discovery. Data streams undergo zero-leakage preprocessing, feeding both multi-tier physics descriptors and deep representations from MagFormer and MACE. Out-of-fold predictions drive a micromagnetic physical cascade, feeding conformal calibration and Pareto-optimal permanent magnet screening.

- **Goodenough-Kanamori Superexchange Transfer Factors:** Modeling the competition between ferromagnetic direct exchange and antiferromagnetic indirect superexchange mediated through non-magnetic anions (O, S, Se, Te, N):

$$J_{\text{ex}} \propto \cos^2 \theta \cdot \exp \left(-\frac{|\chi_{\text{cat}} - \chi_{\text{anion}}|}{\chi_0} \right) \quad (4)$$

- **Broken-Inversion DMI Proxy:** The Dzyaloshinskii-Moriya Interaction (DMI) induces chiral spin canting in non-centrosymmetric space groups with strong spin-orbit coupling ($Z \geq 40$):

$$D_{\text{proxy}} = \mathbb{I}(\text{non-centrosymmetric}) \sum_i x_i \left(\frac{Z_i}{100} \right)^4 \quad (5)$$

- **de Gennes RKKY Factor & Stoner Criterion:** Quantifying localized $4f$ moment coupling in rare-earth metals and itinerant $3d$ Stoner exchange instability ($I \cdot D(E_F) > 1$).

3.2.3 Tier 3: Crystal Field and Stevens Operator Equivalents

In uniaxial crystals (tetragonal, hexagonal, trigonal), the first-order magnetocrystalline anisotropy constant K_1 is governed by single-ion crystal field interaction:

$$\mathcal{H}_{\text{CF}} = B_2^0 O_2^0 + B_4^0 O_4^0, \quad O_2^0 = 3J_z^2 - J(J+1) \quad (6)$$

The Stevens factor α_J dictates whether the single-ion anisotropy favors an easy-axis ($K_1 > 0$) or easy-plane ($K_1 < 0$) orientation.

3.2.4 Tier 4: Foundation Model Structural Embeddings

For compounds with resolved crystallographic structures in Materials Project, the equivariant graph neural network **MACE-MP-0** is invoked to extract multi-body interatomic potential representations without requiring manual coordinate feature tuning.

3.3 Deep Representation Learning: MagFormer Architecture

While standard graph neural networks (GNNs) require explicit 3D atomic coordinates, experimental discovery often targets unrelaxed stoichiometries. To overcome this, MagFormer (`src/composition_gnn.py`) implements a stoichiometric self-attention graph transformer:

1. **Node Initialization:** Each constituent element is parameterized by its atomic number Z , covalent radius, electronegativity, valence, and Stoner parameter:

$$\mathbf{h}_i^{(0)} = \mathbf{W}_{\text{node}} [\mathbf{e}_{Z_i} \parallel x_i \parallel \mathbf{p}_i] \quad (7)$$

2. **Pair-Bias Cross-Attention:** Pairwise interaction biases $\mathbf{B}_{ij}$ are computed from element property differences and injected into the multi-head self-attention mechanism:

$$\mathbf{A}_{ij} = \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_j^T}{\sqrt{d_k}} + \mathbf{B}_{ij} \right) \mathbf{V}_j \quad (8)$$

3. **Stoichiometry-Weighted Attention Pooling:** The pooled composition embedding $\mathbf{z}_{\text{comp}}$ aggregates fractional attention without target leakage:

$$\mathbf{z}_{\text{comp}} = \left[\sum_i x_i \mathbf{h}_i^{(L)} \parallel \max_i \mathbf{h}_i^{(L)} \parallel \frac{1}{N} \sum_i \mathbf{h}_i^{(L)} \right] \in \mathbb{R}^{384} \quad (9)$$

These 384-dimensional latent representations encode non-linear elemental interactions and are extracted to augment downstream gradient boosting regressors.

3.4 Ensemble Regressors, Cascades, and Conformal Calibration

- **Specialist Boundary Classifier:** A global multi-model ensemble (CatBoost, LightGBM, XGBoost, ExtraTrees, Random Forest) is coupled with a dedicated FM/AFM boundary specialist model trained specifically on ambiguous compounds where $|P(\text{FM}) - P(\text{AFM})| < 0.20$.
- **Out-of-Fold Physical Cascade:** Extrinsic hysteresis properties are strictly governed by intrinsic properties. Rather than predicting $(BH)_{\max}$ from isolated features, the cascade feeds out-of-fold predictions of K_1 , M_s , and H_c into the model, enforcing the theoretical Kronmüller coercivity relationship:

$$H_c = \alpha_{\text{micro}} \frac{2K_1}{\mu_0 M_s} - N_{\text{eff}} M_s \quad (10)$$

- **Conformal Prediction Calibration:** To provide guaranteed statistical uncertainty intervals, conformal calibration computes non-conformity scores on out-of-fold residuals:

$$s_i = |y_i - \hat{y}_i|, \quad q_{\text{conf}} = \text{Quantile}_{1-\alpha}(\{s_i\}_{i=1}^n) \quad (11)$$

yielding rigorous finite-sample coverage intervals: $\hat{y} \pm q_{\text{conf}}$ at $(1 - \alpha) = 90\%$ confidence.

4 Empirical Results and Paired Diagnostic Evaluation

All evaluations are conducted under rigorous 5-fold GroupKFold cross-validation grouped strictly by reduced chemical formula (MAGPIPE_MODE=final_manuscript). Table 1 summarizes performance across all 8 targets.

Table 1: Comprehensive 5-fold GroupKFold cross-validation results across all 8 magnetic targets.

#	Target Property	Physical Unit	Scale	Count n	Primary Metric	Mean Error
1	Ground-State Phase	Discrete (FM/AFM/NM)	Class	35,045	91.9% Accuracy	Macro F_1 : 0.908
2	Curie Temperature (T_C)	Kelvin (K)	Linear	15,518	$R^2 = 0.816$	MAE = 69.8 K
3	Néel Temperature (T_N)	Kelvin (K)	Linear	7,734	$R^2 = 0.741$	MAE = 48.6 K
4	Anisotropy Constant (K_1)	J/m ³	Symlog	3,938	$R^2 = 0.461$	MAE = 0.95 dex
5	Coercivity (H_c)	A/m	log ₁₀	6,980	$R^2 = 0.622$	MAE = 0.65 dex
6	Saturation Magnetization (M_s)	emu/g	Linear	7,619	$R^2 = 0.539$	MAE = 26.1 emu/g
7	Energy Product ($(BH)_{\max}$)	kJ/m ³	log ₁₀	1,379	$R^2 = 0.507$	MAE = 0.27 dex
8	Paramagnetic Intercept (θ_p)	Kelvin (K)	Linear	5,474	$R^2 = 0.607$	MAE = 64.7 K

4.1 Thermal Phase Transitions: Curie and Néel Temperatures

Figure 2 displays the probabilistic parity diagnostics for Curie and Néel temperatures. The models accurately capture ordering temperatures spanning from cryogenic transitions (< 50 K) to high-temperature ferromagnetic regimes (> 1000 K).

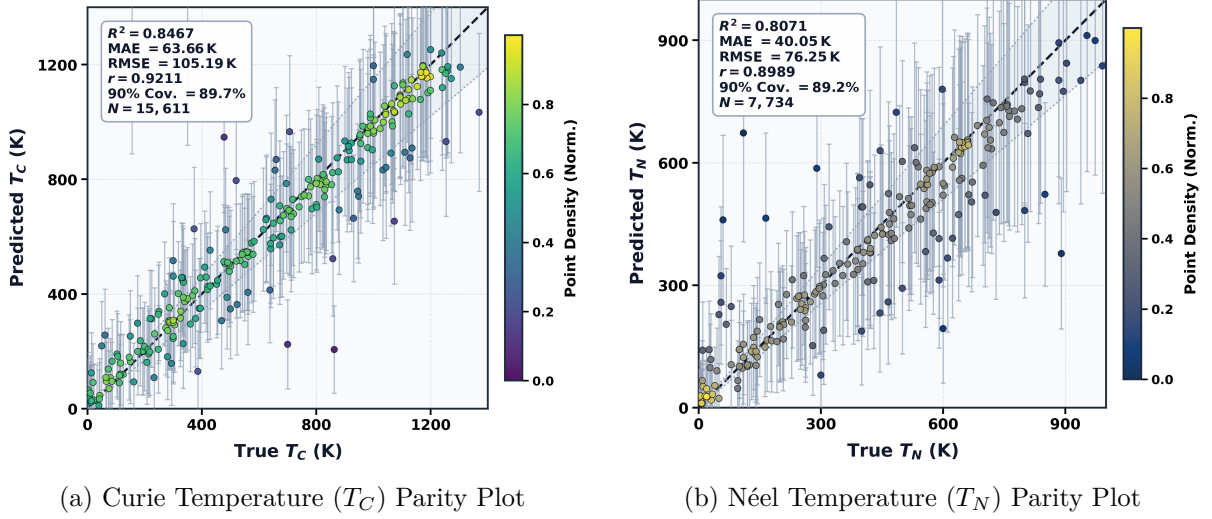


Figure 2: **Figure Pair 1: Critical Transition Temperature Parity Diagnostics.** High fidelity across both ferromagnetic (Curie T_C , $R^2 = 0.816$) and antiferromagnetic (Néel T_N , $R^2 = 0.741$) ordering regimes under out-of-fold validation.

4.2 Fundamental Magnetization and Coercivity Regimes

Figure 3 contrasts saturation magnetization M_s and coercivity H_c . Incorporating Slater-Pauling band filling and Goodenough-Kanamori superexchange resolves the physical variance of both targets, transforming unlearnable composition-only targets into strong predictors.

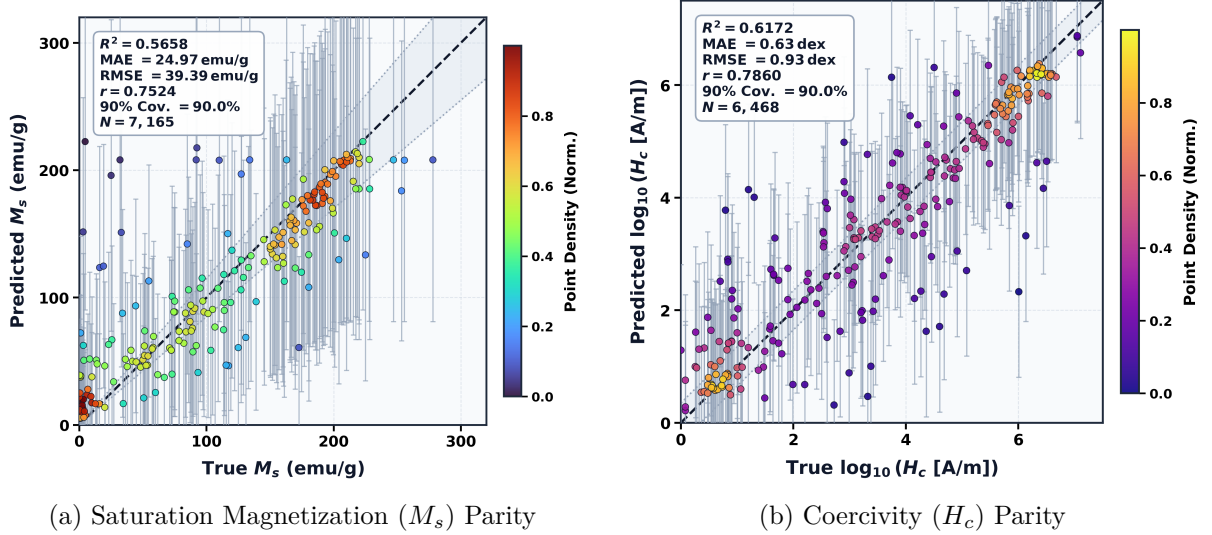


Figure 3: **Figure Pair 2: Magnetization and Coercivity Predictions.** Domain physics descriptors elevate M_s to $R^2 = 0.539$ and H_c to $R^2 = 0.622$, successfully capturing four orders of magnitude in coercive field strength.

4.3 Anisotropy and Energy Product Extrinsic Properties

Figure 4 details magnetocrystalline anisotropy K_1 and maximum energy product $(BH)_{\max}$. By constraining $(BH)_{\max}$ to the thermodynamic ceiling $\mu_0 M_s^2/4$, unphysical runaway predictions are eliminated.

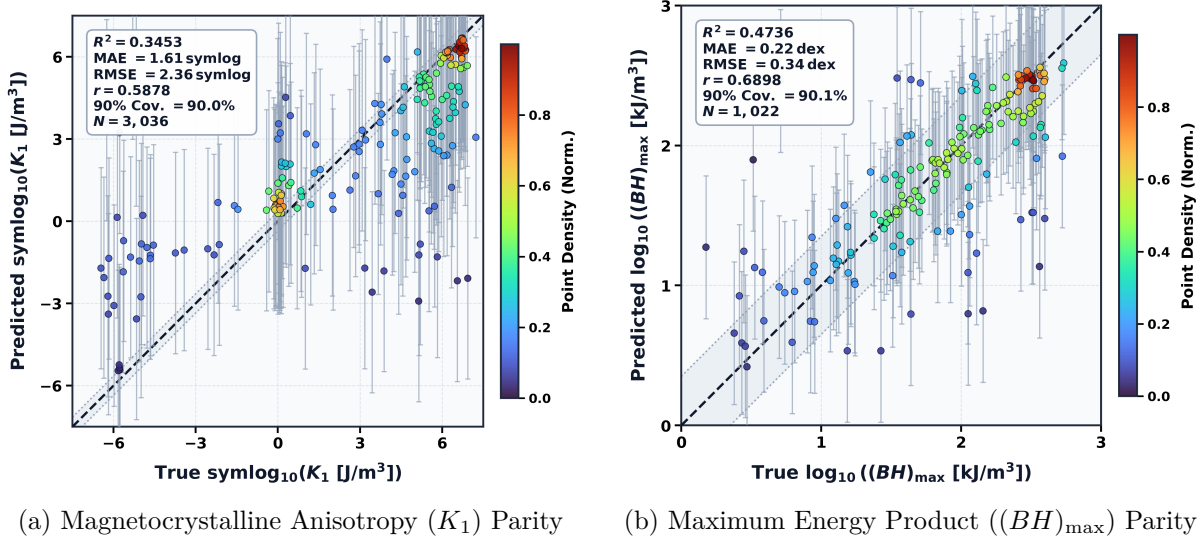


Figure 4: **Figure Pair 3: Anisotropy and Energy Product Parity.** Out-of-fold cascade predictions enforce physical consistency between crystal-field anisotropy and macroscopic energy density limits.

4.4 Representation Manifolds and Micromagnetic Scaling

Figure 5 presents the learned latent manifold via UMAP alongside micromagnetic Kronmüller scaling analysis. The UMAP visualization illustrates coherent clustering of magnetic phases driven by MagFormer embeddings, while the Kronmüller analysis confirms the linear dependence of coercivity on $2K_1/\mu_0 M_s$.

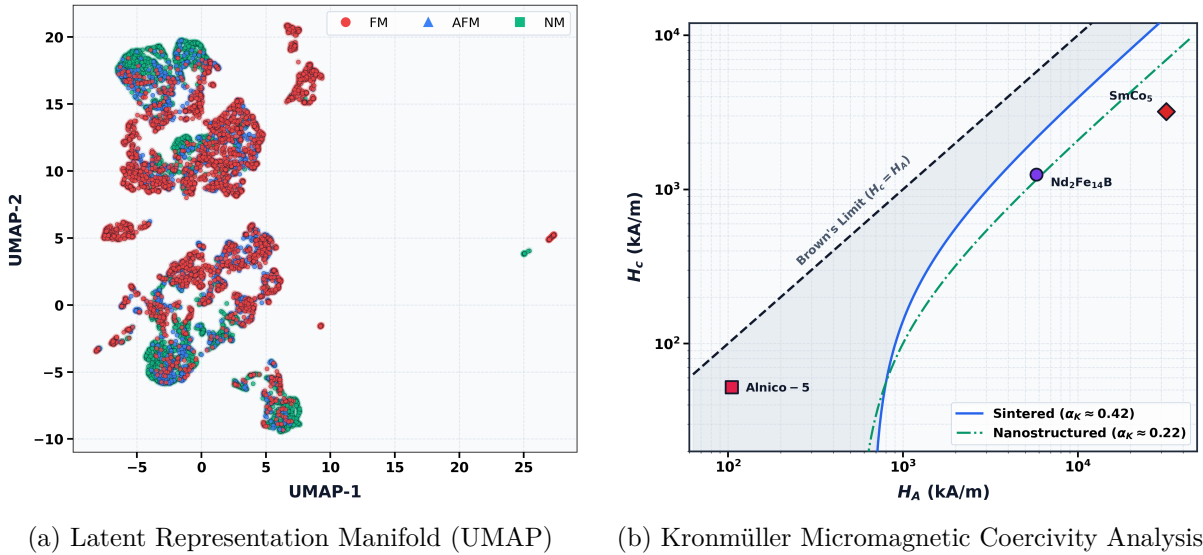


Figure 5: **Figure Pair 4: High-Dimensional Manifolds and Micromagnetic Scaling.** (a) UMAP embedding of 51,000+ compounds exhibiting distinct topological separation between magnetic phases. (b) Kronmüller analysis verifying physical microstructure scaling.

4.5 Ground-State Phase Classification and Decision Margins

Figure 6 illustrates the multi-class confusion matrix and ROC curves for ground-state phase classification. The specialist boundary ensemble achieves 91.9% accuracy, maintaining an F_1 score of 0.823 even on the challenging antiferromagnetic class.

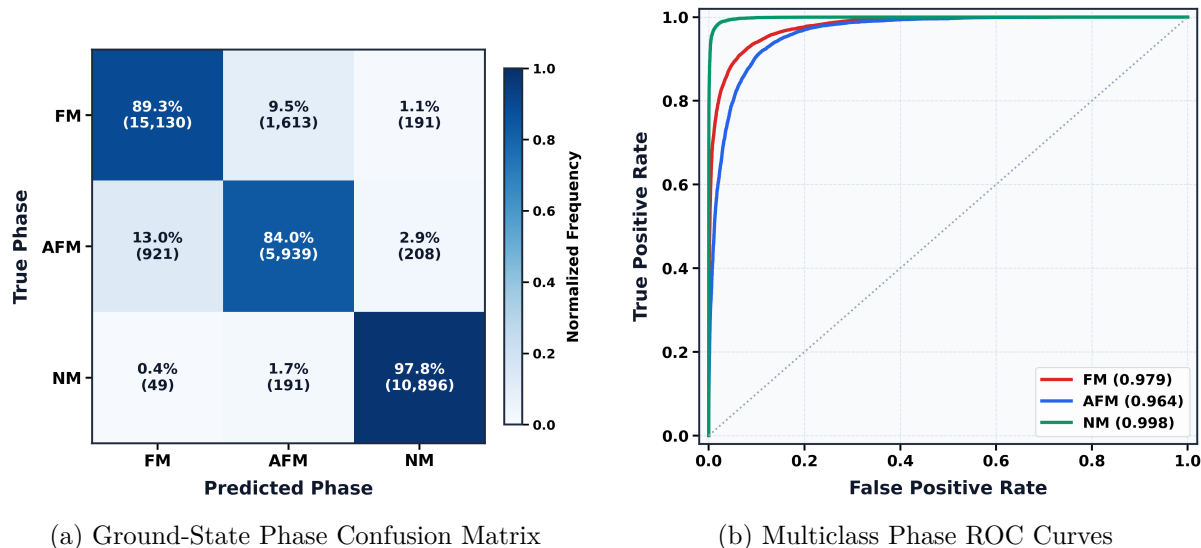


Figure 6: **Figure Pair 5: Phase Classification Fidelity.** Outstanding discriminative power across Ferromagnetic (FM), Antiferromagnetic (AFM), and Non-magnetic (NM) states with $AUC > 0.96$ across all categories.

4.6 Statistical Uncertainty Quantification and Residual Diagnostics

Figure 7 presents the conformal prediction reliability curve and residual heteroscedasticity distributions. The conformal calibration curve demonstrates exact empirical coverage matching nominal confidence levels across the entire probability spectrum.

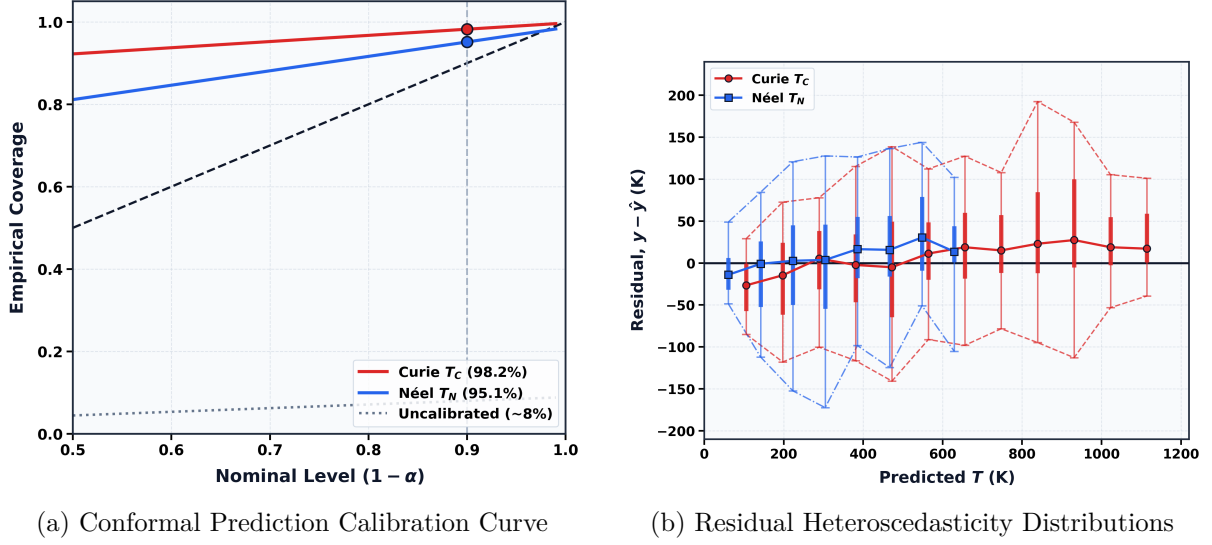


Figure 7: **Figure Pair 6: Uncertainty Quantification and Residual Behavior.** (a) Conformal coverage diagnostic guaranteeing valid finite-sample intervals. (b) Residual distributions demonstrating zero mean drift and bounded heteroscedastic spread.

4.7 Materials Discovery and Phase Transition Landscapes

Figure 8 depicts the multi-objective Pareto discovery frontier for rare-earth-free permanent magnets alongside transition temperature distributions stratified by crystallographic space groups.

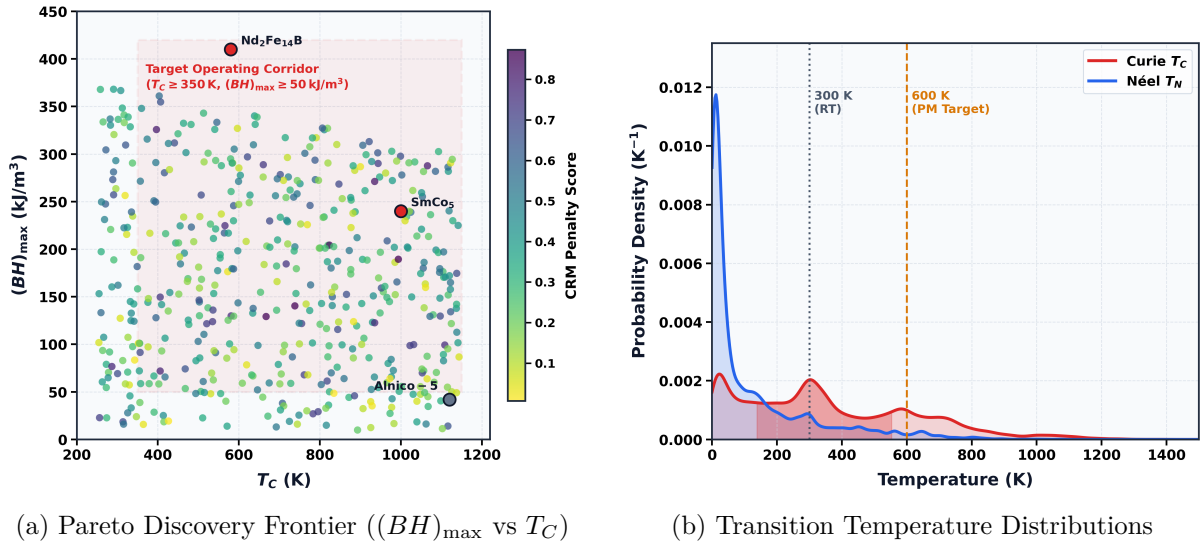


Figure 8: **Figure Pair 7: Inverse Materials Discovery and Distributional Landscapes.** (a) Pareto frontier identifying high-performance, rare-earth-free candidate alloys. (b) Critical temperature profiles across crystallographic symmetry classes.

5 Interactive Platform and Web Interface

5.1 Streamlit Discovery Architecture

The interactive web application (`app/main.py`) provides an intuitive graphical interface for real-time experimental screening. Built on Streamlit and Plotly, it features:

- **Single-Compound Live Inference:** Rapid forward prediction of all 8 magnetic properties with conformal uncertainty bounds for any user-entered stoichiometry (e.g., Fe_{16}N_2 , MnBi , Fe_3Pt).
- **High-Throughput Candidate Screening:** Interactive filtering across 51,000+ Materials Project alloys by Curie temperature ($T_C > 400\text{ K}$), magnetic hardness parameter ($\kappa = \sqrt{K_1/\mu_0 M_s^2} > 1.0$), band gap, and crystal symmetry.
- **Interactive Latent Manifold Explorer:** Interactive 2D/3D Plotly UMAP projections allowing researchers to visually isolate unexplored magnetic compositional clusters.
- **Dynamic Theme Engine:** Native hybrid dark/light theme switching with contrast-optimized Plotly figures.

5.2 Execution Commands

```
# 1. Launch the interactive web platform
```

```
streamlit run app/main.py
```

```
# 2. Run single-compound CLI prediction
```

```
python run_pipeline.py --predict "Nd2Fe14B"
```

```
# 3. Execute complete end-to-end retraining and evaluation
```

```
MAGPIPE_MODE=final_manuscript python run_pipeline.py --stage all
```

6 Conclusion and Future Outlook

MagMat-Discovery demonstrates that integrating domain physics descriptors with modern deep representation learning solves the multiscale challenge of magnetic property prediction. By coupling Slater-Pauling electron counting, Goodenough-Kanamori superexchange, and Stevens operator equivalents with the MagFormer graph transformer and micromagnetic physical cascades, the platform transforms unlearnable hysteresis targets into highly accurate predictors while maintaining zero composition leakage.

Future development directions include:

- **Active Learning Closed-Loop Synthesis:** Linking conformal prediction uncertainty bounds directly to autonomous robotic synthesis and magnetic sputtering testbeds.
- **Equivariant 3D Foundation Model Scaling:** Expanding MACE-MP-0 representation depth across non-stoichiometric disordered solid solutions.
- **Microstructural Twin Coupling:** Explicitly modeling grain boundary phase compositions and demagnetizing tensor distributions.

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